Wei Zhang, Ph.D.

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Division of Biostatistics, Department of Public Health Sciences
University of Miami Miller School of Medicine, Miami, FL 33136

Research Interests

- Statistical machine learning methods for integrative multi-omics analysis and representation learning
- Method development for biomarker discovery and disease subtyping in neurodegenerative diseases and cancer
- Interpretable and scalable models for high-dimensional genomic, epigenomic, and neuroimaging data

Education

University of Miami *Miami, FL*
Ph.D. in Biostatistics (Advisor: X. Steven Chen, Ph.D.) Aug 2024

- Dissertation: *Integrative Multi-Omics Analysis Using Multivariate Random Forest*

The George Washington University *Washington, DC*
M.S. in Statistics May 2019

State University of New York at Binghamton *Binghamton, NY*
B.S. in Economic Analysis & Actuarial Mathematics May 2017

Research Positions

University of Miami, Miller School of Medicine *Miami, FL*
Postdoctoral Associate, Dept. of Public Health Sciences (Mentors: Lily Wang, Ph.D.; X. Steven Chen, Ph.D.) Sep 2024 – Present

- Develop statistical machine learning methods for multi-omics integration and multimodal aging biomarkers, with applications to Alzheimer's disease and cancer.

Graduate Research Assistant, Dept. of Public Health Sciences May 2022 – Aug 2024

- Conducted integrative genomic and epigenomic analyses in Alzheimer's and cancer cohorts, developing transcriptome-based prediction models and biomarker selection methods.

Publications & Preprints

[*] *Corresponding author. Full list: Google Scholar*

Peer-Reviewed Journal Articles

1. O'Shea DM*, Wang L, Lukacsovich D, Dhanekula D, **Zhang W**, Galvin C, Joshi M, Besser L, Rundek T, Galvin JE. (2026). "Development and Validation of MethylCog, a Blood DNA Methylation Proxy for Cognition." *Alzheimer's & Dementia*, 22(5):e71421. doi:10.1002/alz.71421 [2025 JIF: 12.8]
2. Lukacsovich D, Young JI, Gomez L, Kunkle BW, Mao Z, **Zhang W**, Chen XS, O'Shea DM, Rundek T, Martin ER, Wang L*, for the Alzheimer's Disease Neuroimaging Initiative. (2026). "Blood DNA Methylation Signature of Cognitive Reserve Moderates the Association between CSF Tau Pathology and Memory in Prodromal Alzheimer's Disease." *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 12(2):e70257. doi:10.1002/trc2.70257 [2025 JIF: 7.4]
3. **Zhang W**, Lukacsovich D, Young JI, Gomez L, Schmidt MA, Kunkle BW, Chen X, Martin ER,

Wang L*, for the Alzheimer's Disease Neuroimaging Initiative. (2026). "The Aging Epigenome: Integrative Analyses Reveal Intersection with Alzheimer's Disease." *GeroScience*, 48:3185–3203. doi:10.1007/s11357-026-02195-x [2025 JIF: 6.0]

4. Lukacsovich D, Young JI, Gomez L, Schmidt MA, **Zhang W**, Kunkle BW, Chen XS, Martin ER, Wang L*. (2026). "From Aging to Alzheimer's Disease: Concordant Brain DNA Methylation Changes in Late Life." *Genome Medicine*. doi:10.1186/s13073-026-01698-8 [2025 JIF: 10.8]
5. **Zhang W***, Huang H, Wang L, Lehmann BD, Chen XS*. (2026). "An Integrative Multiomics Random Forest Framework for Robust Biomarker Discovery." *GigaScience*, 15:giaf148. doi:10.1093/gigascience/giaf148 [2025 JIF: 5.0]
6. **Zhang W**, Lukacsovich D, Young JI, Gomez L, Schmidt MA, Martin ER, Kunkle BW, Chen XS, O'Shea DM*, Galvin JE*, Wang L*. (2025). "DNA Methylation Signature of a Lifestyle-based Resilience Index for Cognitive Health." *Alzheimer's Research & Therapy*, 17:88. doi:10.1186/s13195-025-01733-7 [2025 JIF: 8.9]
7. **Zhang W**, Young JI, Gomez L, Schmidt MA, Lukacsovich D, Kunkle B, Chen XS, Martin ER, Wang L*. (2025). "Blood DNA Methylation Signature for Incident Dementia: Evidence from Longitudinal Cohorts." *Alzheimer's & Dementia*, 21:e14496. doi:10.1002/alz.14496 [2025 JIF: 12.8]
8. **Zhang W**, Wu C, Huang H, Bleu P, Zambare W, Alvarez J, Wang L, Paty PB, Romesser PB, Smith JJ*, Chen XS*. (2025). "Enhancing Chemotherapy Response Prediction via Matched Colorectal Tumor-Organoid Gene Expression Analysis and Network-Based Biomarker Selection." *Translational Oncology*, 52:102238. doi:10.1016/j.tranon.2024.102238 [2025 JIF: 4.9]
9. Lukacsovich D, O'Shea D, Huang H, **Zhang W**, Young JI, Chen XS, Dietrich S-T, Kunkle B, Martin ER, Wang L*. (2024). "MIAMI-AD (Methylation in Aging and Methylation in AD): An Integrative Knowledgebase That Facilitates Explorations of DNA Methylation Across Sex, Aging, and Alzheimer's Disease." *Database*, 2024:baae061. doi:10.1093/database/baae061 [2025 JIF: 2.7]
10. **Zhang W**, Young JI, Gomez L, Schmidt MA, Lukacsovich D, Varma A, Chen XS, Kunkle B, Martin ER, Wang L*. (2024). "Critical Evaluation of the Reliability of DNA Methylation Probes on the Illumina MethylationEPIC v1.0 BeadChip Microarrays." *Epigenetics*, 19(1):2333660. doi:10.1080/15592294.2024.2333660 [2025 JIF: 3.7]
11. **Zhang W**, Young JI, Gomez L, Schmidt MA, Lukacsovich D, Varma A, Chen XS, Martin ER, Wang L*. (2023). "Distinct CSF Biomarker-Associated DNA Methylation in Alzheimer's Disease and Cognitively Normal Subjects." *Alzheimer's Research & Therapy*, 15:78. doi:10.1186/s13195-023-01216-7 [2025 JIF: 8.9]
12. **Zhang W**, Li E, Wang L, Lehmann BD, Chen XS*. (2023). "Transcriptome Meta-Analysis of Triple-Negative Breast Cancer Response to Neoadjuvant Chemotherapy." *Cancers*, 15(8):2194. doi:10.3390/cancers15082194 [2025 JIF: 4.8]
13. Silva TC, **Zhang W**, Young JI, Gomez L, Schmidt MA, Varma A, Chen XS, Wang L*. (2022). "Distinct Sex-Specific DNA Methylation Differences in Alzheimer's Disease." *Alzheimer's Research & Therapy*, 14(1):121. doi:10.1186/s13195-022-01070-z [2025 JIF: 8.9]

Preprints

- **Zhang W***, Wang L, Franzmann EJ, Chen XS. (2026). "Multivariate Random Forests for Cross-Modal Multi-Omics Integration." *bioRxiv*, 2026.06.17.732933. doi:10.64898/2026.06.17.732933
- Chen XS, Lukacsovich D, Zambare W, Wu C, Huang H, **Zhang W**, Kim MJ, et al. (2025). "Integrating Tumor and Organoid DNA Methylation Profiles Reveals Robust Predictors of Chemotherapy Response in Rectal Cancer." *medRxiv*, 2025.02.28.25322951. doi:10.1101/2025.02.28.25322951

Grant Applications

NIH/NINDS K99/R00 Pathway to Independence Award Submitted Oct 2025

Role: PI · Mentor: Lily Wang, Ph.D. · Co-Mentors: Tatjana Rundek, M.D., Ph.D.; David Loewenstein, Ph.D. · Not Funded (Competitive, Not Discussed)

Criterion scores (1–9, 1 = best):	Candidate	Career Dev.	Research Plan	Mentors	Environment
Reviewers 1 / 2 / 3:	1 / 1 / 1	3 / 3 / 2	3 / 4 / 5	3 / 3 / 2	1 / 1 / 1

Alzheimer's Association Research Fellowship for All (AARFA) Submitted Oct 2025

Role: PI · Mentor: Lily Wang, Ph.D. · Co-Mentors: Tatjana Rundek, M.D., Ph.D.; David Loewenstein, Ph.D. · Not Funded

Reviewer scores (1–9 scale, 1 = best): 1 (Exceptional) · 2 (Outstanding) · 3 (Excellent)

Software & Open-Source Tools

- **multiRF**: An R package for integrating matched multi-omics datasets with multivariate random forests. It fits directed forest models across omics blocks, learns sample-by-sample similarity from shared terminal-node structure, and decomposes the result into shared and omics-specific components for clustering, variable selection, and visualization. [GitHub]

Presentations

O = Oral; P = Poster; Presenting author

[**O.1**] **Zhang W** and Chen XS. "Multivariate Random Forest-Based Clustering for Integrative Multi-Omics Analysis." *ASA Florida Chapter Annual Meeting*, Mar 2026. Miami, FL. Contributed Talk.

[**O.2**] **Zhang W** and Chen XS. "An Integrative Multi-Omics Random Forest Framework for Robust Biomarker Discovery." *STATGEN*, May 2025. Minneapolis, MN. Contributed Talk.

[**O.3**] **Zhang W** and Chen XS. "Unlocking the Potential of Multi-Omics Data Integration Using Multivariate Random Forest Approach." *ENAR Annual Meeting*, Mar 2024. Baltimore, MD. Contributed Talk.

[**P.1**] **Zhang W**, et al. "Multivariate Random Forest-Based Clustering for Integrative Multi-Omics Analysis." *Joint Statistical Meetings (JSM)*, Aug 2026. Boston, MA. Poster.

[**P.2**] **Zhang W**, Rundek T, Loewenstein D, Wang L. "A Disease-Focused Integrative Epigenetic and Neurostructural Age Clock for Alzheimer's Risk." *Alzheimer's Association International Conference (AAIC)*, Jul 2026. Virtual. Poster.

[**P.3**] Wang L, **Zhang W**, Gomez L, Kunkle BW, Lukacsovich D, Schmidt MA, Varma A, Griswold AJ, Bush WS, Martin ER. "An X Chromosome-Wide DNA Methylation Study of Alzheimer's Disease." *AAIC*, Jul 2024.

[**P.4**] Wang L, **Zhang W**, Young JI, Gomez L, Schmidt MA, Lukacsovich D, Varma A, Chen XS, Martin ER. "Distinct CSF Biomarker-Associated DNA Methylation in Alzheimer's Disease and Cognitively Normal Subjects." *AAIC*, Jul 2023.

[**P.5**] **Zhang W** and Chen XS. "Iterative Multivariate Random Forest for Feature Selection in Integrating Multi-Omics Datasets." *ASA Florida Chapter Annual Meeting*, Mar 2023. Gainesville, FL.

Teaching Experience

EPH 705: Advanced Statistical Methods II *University of Miami*
Teaching Assistant (Instructor: Lily Wang, Ph.D.) 2022 – 2024

- Led weekly office hours and Q&A sessions; prepared supplementary materials and worked one-on-one with students to reinforce course concepts.

STAT 6214: Applied Linear Models *The George Washington University*
Teaching Assistant (Instructor: Emre Barut, Ph.D.) 2018

Honors & Awards

- **Award of Academic Merit**, University of Miami Aug 2024
- **Best Student Poster Award**, ASA Florida Chapter Meeting Mar 2023
- **Travel Award**, University of Miami Mar 2023

Professional Development

- **Duke Electronic Health Records Study Design Workshop**, Duke University Dec 2024
- **Code Rigor and Reproducibility with R Boot Camp**, Columbia University Jul 2023

Professional Service

Peer Review

- Nature Communications, Neurobiology of Aging, Statistics in Medicine, Scientific Reports, Discover Applied Sciences, Biology Direct, Discover Oncology, Medicine in Omics

Other Service

- Reviewer, ASA South Florida Student Data Challenge (2026)

Professional Memberships

- International Biometric Society, Eastern North American Region (ENAR)
- American Statistical Association (ASA)
- International Society to Advance Alzheimer's Research and Treatment (ISTAART)

Technical Skills

- **Programming:** R, Python, SAS, SQL
- **Methods:** Machine learning, random forests, multi-omics integration, deep learning
- **Infrastructure:** Linux, HPC, Git
- **Bioinformatics:** DNA methylation, RNA-seq, EWAS, GWAS

Languages

English (professional), Mandarin Chinese (native), Cantonese (fluent)